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DT081A – Signal and Image Processing

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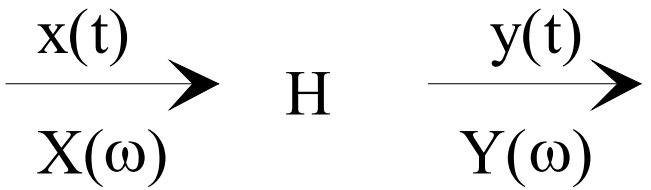


Lecture 7
Filter Design – Spectrum

System Frequency Response

Continuous Time LTI System in Frequency domain

- Continuous Time LTI System

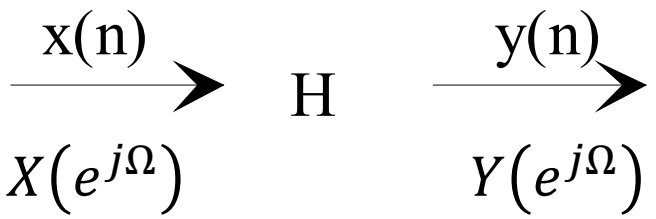


$$Y(\omega) = H(\omega) \cdot X(\omega)$$

- Frequency response $H(\omega) = |H(\omega)| \cdot e^{j \arg H(\omega)}$
- Amplification (Magnitude) $|H(\omega)|$
- Phase shift $\arg H(\omega)$
- Impulse response $h(t) = F^{-1}\{H(\omega)\}$

Discrete Time LTI System in Frequency domain

- Discrete Time LTI System



$$Y(e^{j\Omega}) = H(e^{j\Omega}) \cdot X(e^{j\Omega})$$

- Frequency response $H(e^{j\Omega}) = |H(e^{j\Omega})| \cdot e^{j \arg H(e^{j\Omega})}$
- Amplification (Magnitude) $|H(e^{j\Omega})|$
- Phase shift $\arg H(e^{j\Omega})$
- Impulse response $h(n) = F^{-1}\{H(e^{j\Omega})\}$

Consequences of the System Frequency Response

- Frequency Response $H(\omega)$
  - Decides the frequency characteristics for the LTI system

$$Y(\omega) = H(\omega) \cdot X(\omega)$$

$$|Y(\omega)| \cdot e^{j \arg Y(\omega)} = |H(\omega)| \cdot e^{j \arg H(\omega)} \cdot |X(\omega)| \cdot e^{j \arg X(\omega)}$$

- Amplitude
  - $|Y(\omega)| = |H(\omega)| \cdot |X(\omega)|$
  - $|H(\omega)|$ decides the amplification / damping of the input signal

- Phase
  - $\arg Y(\omega) = \arg H(\omega) + \arg X(\omega)$
  - $\arg H(\omega)$ decides the phase shift of the input signal

Consequences of the System Frequency Response

- Output is given by input and frequency response

$$Y(\omega) = H(\omega) \cdot X(\omega)$$

- An LTI system is *frequency preserving*
  - I.e., Output contains *the same frequencies* as there is in the input
  - The contribution of these frequencies may (in theory) be infinitely damped (amplitude = 0)

- An LTI-system affects only the input signal's
  - Amplitude, and
  - Phase

Filter

Filter

- A filter is a system – commonly an LTI system – for which
  - Amplification / Damping
  - Phase shift (or its derivative – phase runtime, group runtime)

have *consciously been selected for a particular purpose (objective)*

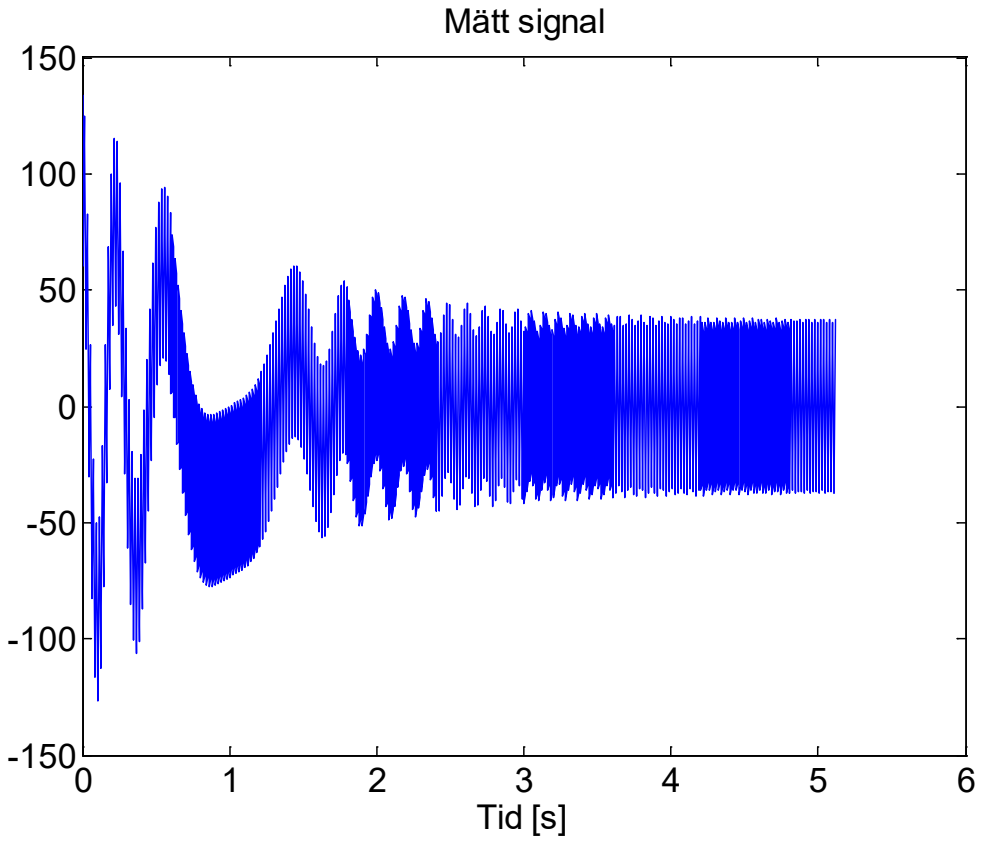


The Principle of Filtering

- In filtering, interesting information is extracted by selecting interesting frequencies.

Example

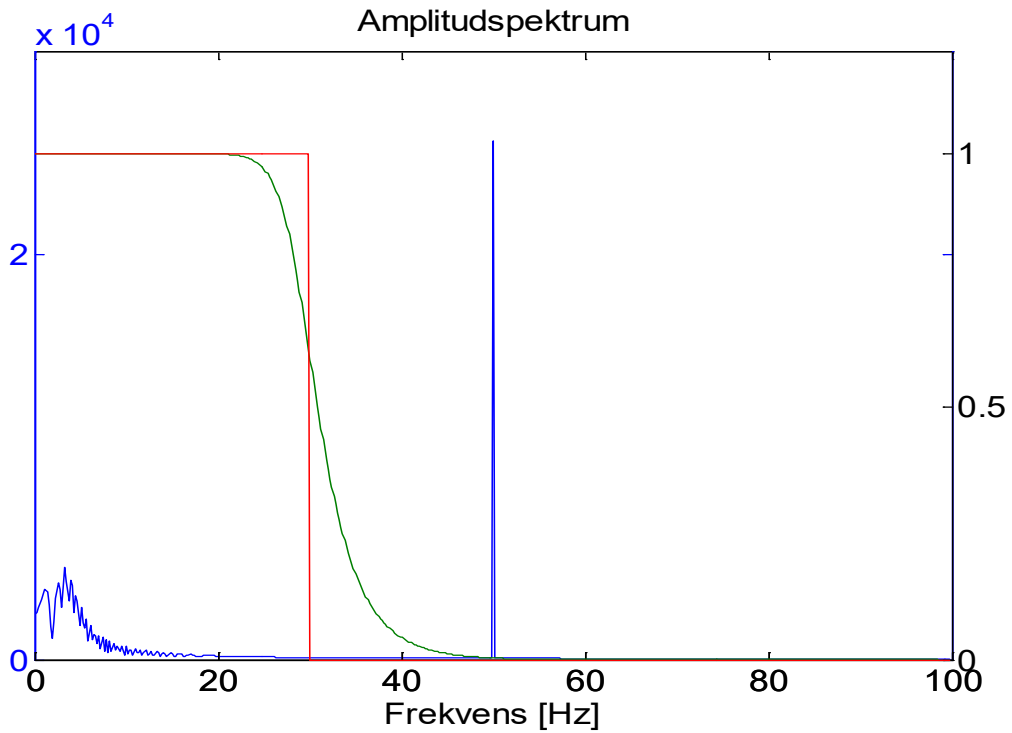
- Measured signal with 50 Hz "hum" from the power source.



The Principle of Filtering

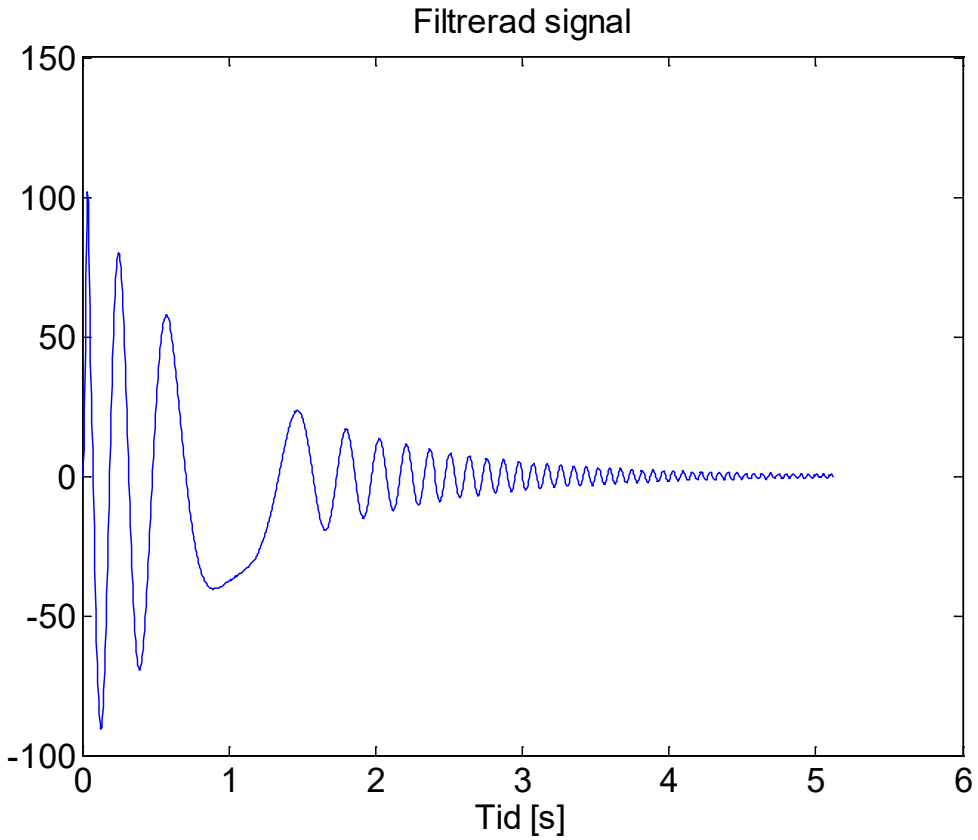
- The spectrum of the signal
  - The frequency content of the signal

- Low pass filter
  - Red: ideal filter
  - Green: real filter



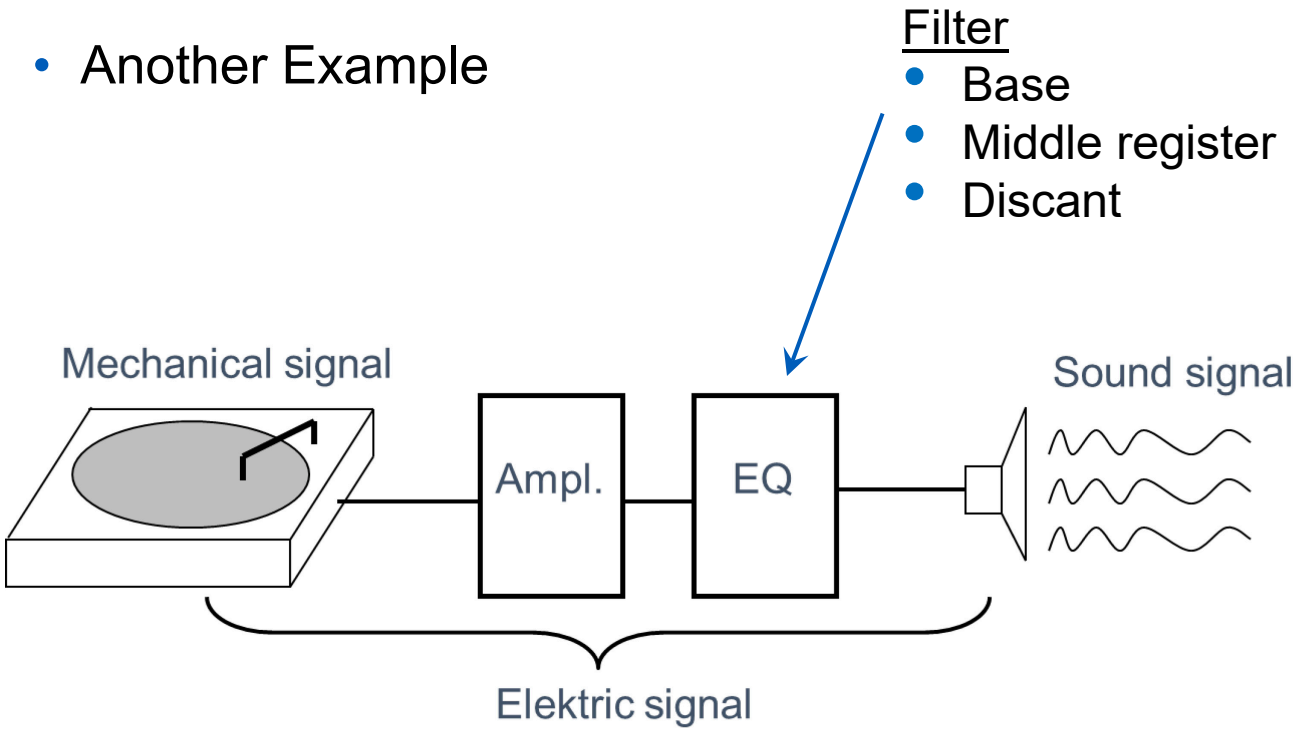
The Principle of Filtering

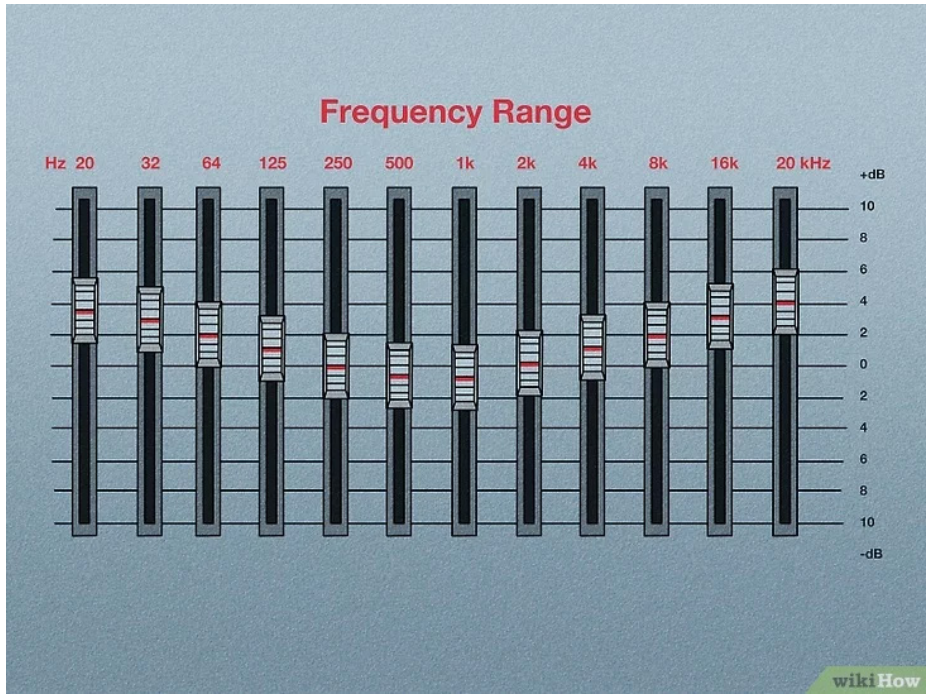
- Filtered signal



The Principle of Filtering

- Another Example





https://www.wikihow.com/Use-a-Graphic-Equalizer

Ideal Filters

The Principle of Filtering

- Filtrering can be carried out (studied) through

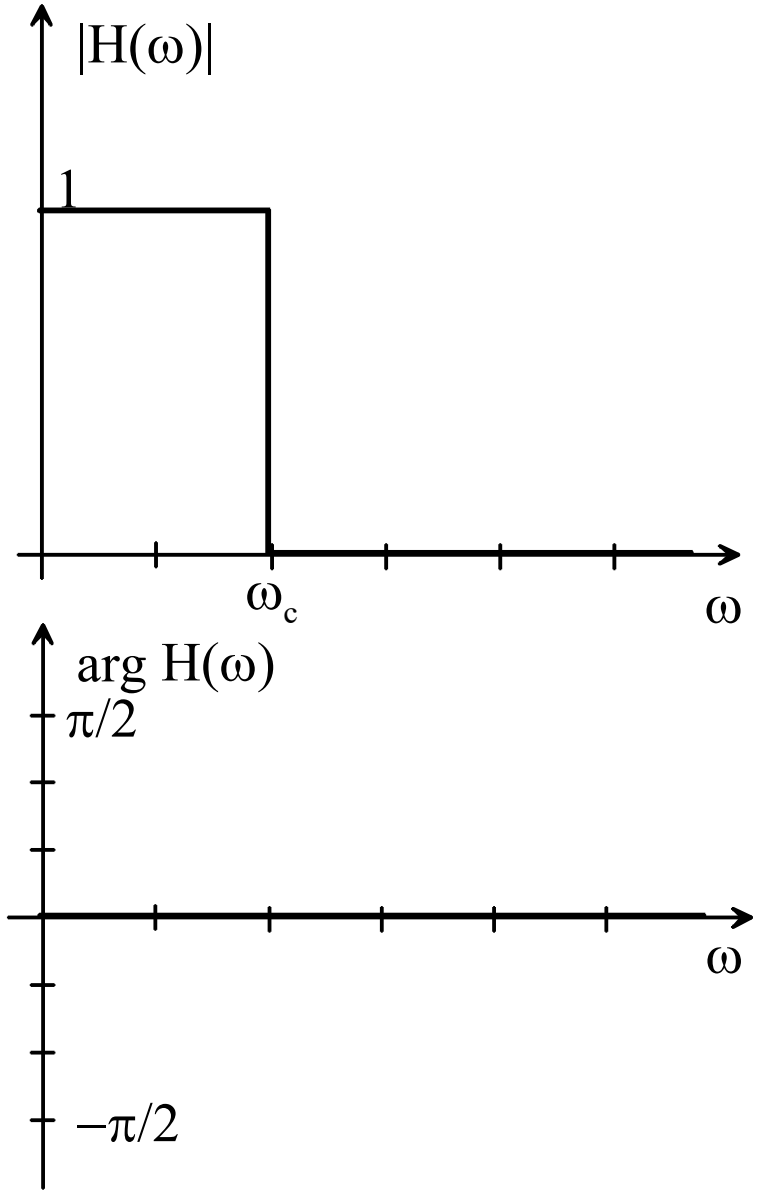
- Filtrering in Frequency domain
  - $x(t) \rightarrow X(\omega)$
  - $Y(\omega) = H(\omega) \cdot X(\omega) \rightarrow y(t)$
  - $Y(\omega) \rightarrow y(t)$

- Filtering in Time domain
  - $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau$

Ideal Filter

- An **Ideal filter** is
  - A *Causal LTI System*

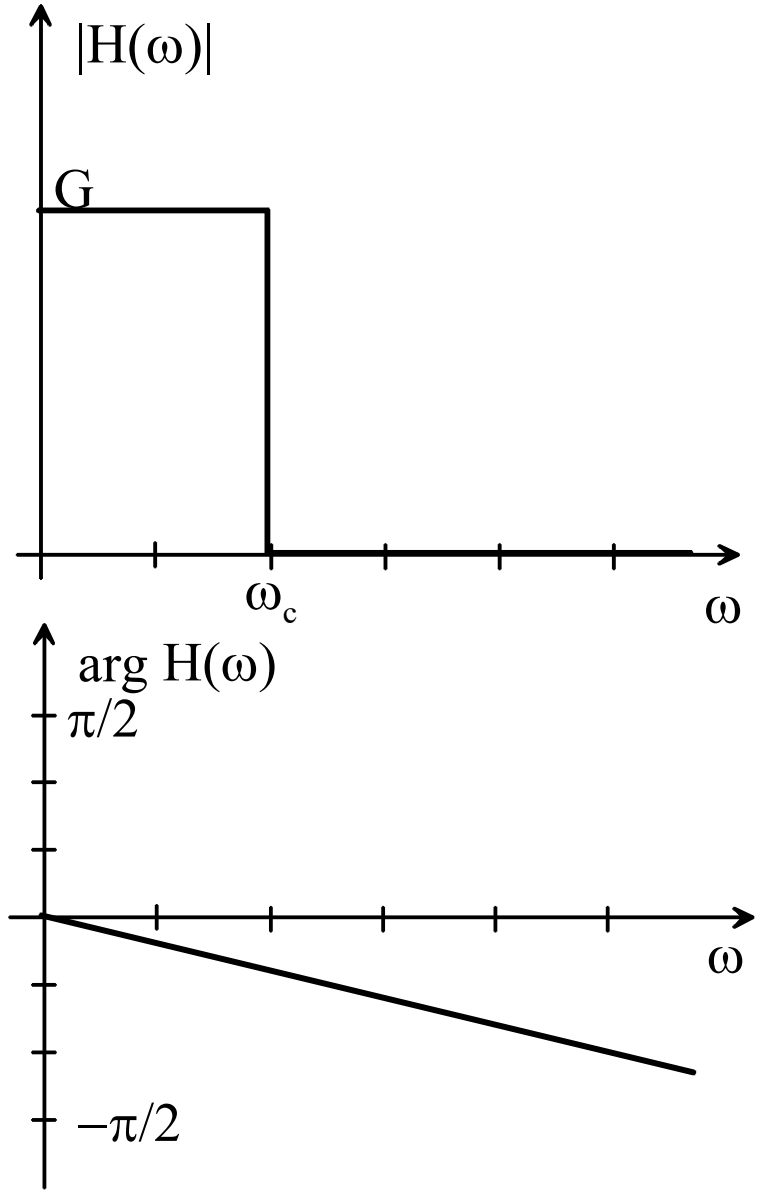
    with
- **Pass band** transmits
  - *All signal components without distortion* – neither damping nor phase shift
- **Stop band** transmits
  - No signal components
  - I.e. It *damps* all signal components *infinitely*



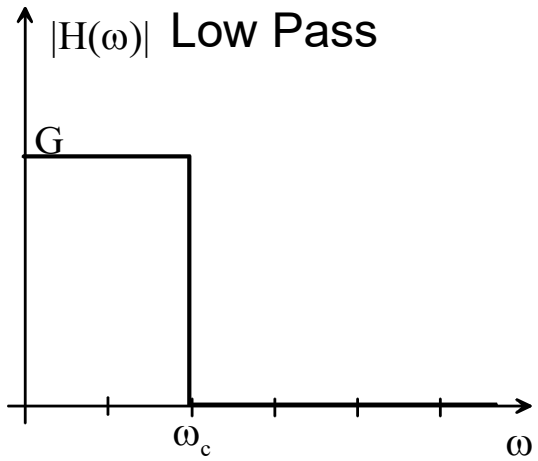
Ideal Filter

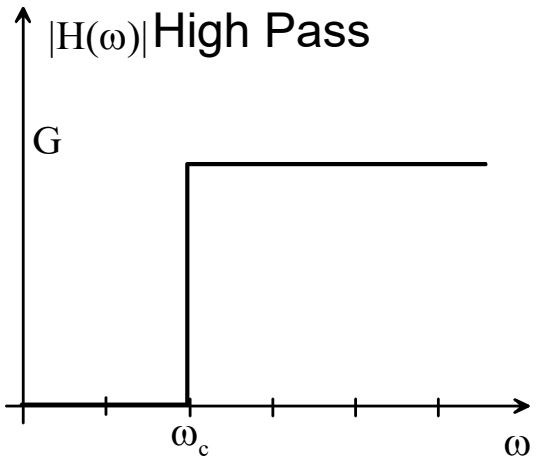
- An **Ideal filter** is
  - A *Causal LTI System*

    that is *distortion free:*
- **Pass band** has
  - *Constant amplification*
  - *Constant time delay* (linear phase shift)
- **Stop band**
  - *damps infinitely*

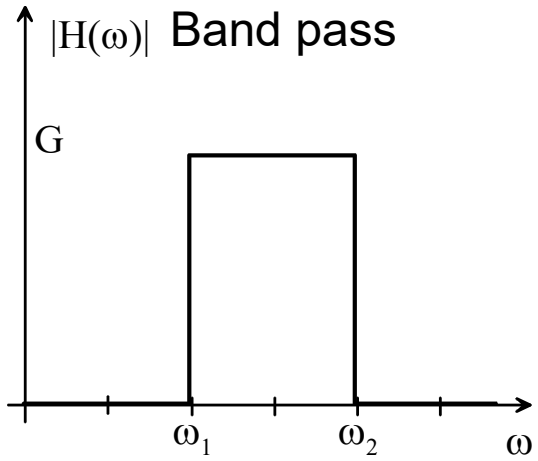


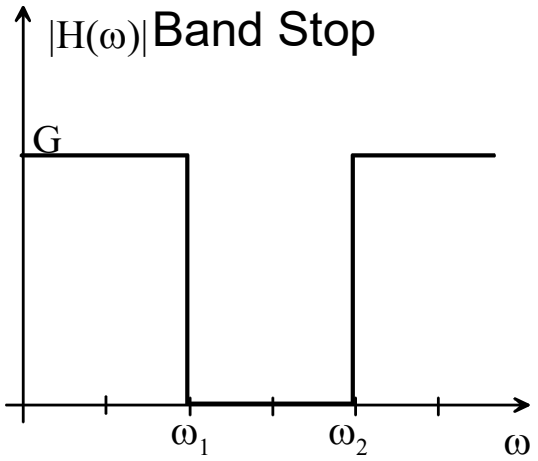
Filter Classification depending on Frequency

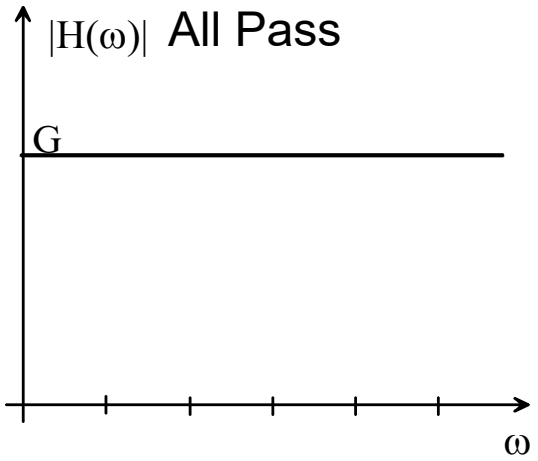




All pass filter is used for
phase compensation

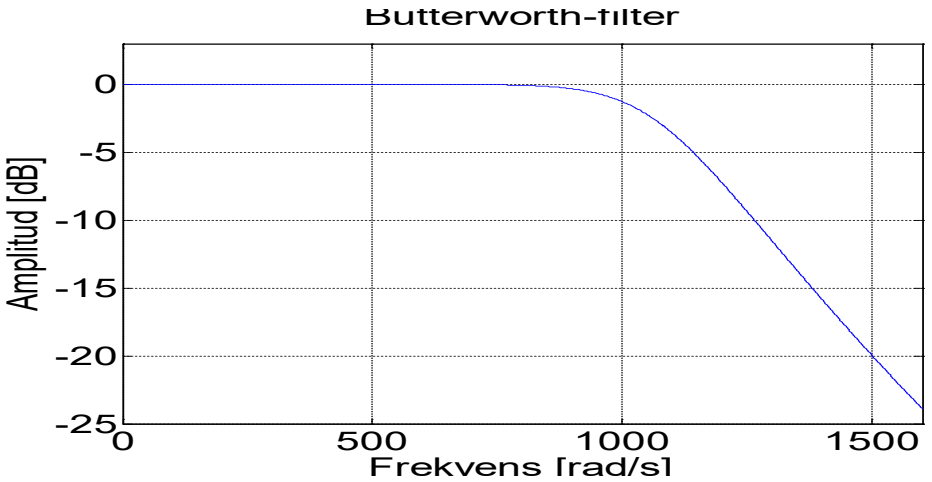


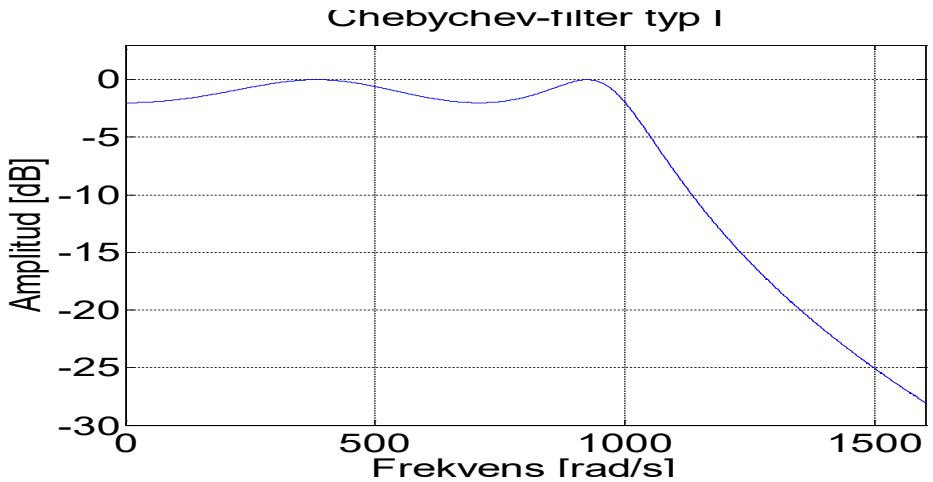




Realisation of Ideal Filter

- Analogue ideal filter is impossible to implement
- Damping cannot be infinite
  - It would not be frequency preserving
- It would imply a non-causal impulse response
- Approximations are used in reality
  - Butterworth
  - Chebychev typ I, typ II
  - Elliptiskt (Cauer)
  - Bessel





Digital Filter

Digital Filter



- Antialiasing Filter
  - Assures no aliasing by removing all frequencies above half the sampling frequency $f_s/2$
- A/D Converter
  - Results in a binary-coded sequence
- Discrete Time Filter
  - Designed according to requirements: LP, HP, BP, BS
- D/A Converter
  - Result in a discrete-amplitude continuous-time signal
- Reconstruction Filter
  - Smoothes the signal by removing frequencies higher than half the sampling frequency $f_s/2$

Advantages and Drawback with Digital Filters

Advantages

- Arbitrarily tight tolerances
- Almost unaffected by external factors
  - (e.g. temp)
- Can be reproduced exactly
- Easy to change
  - (if new needed)

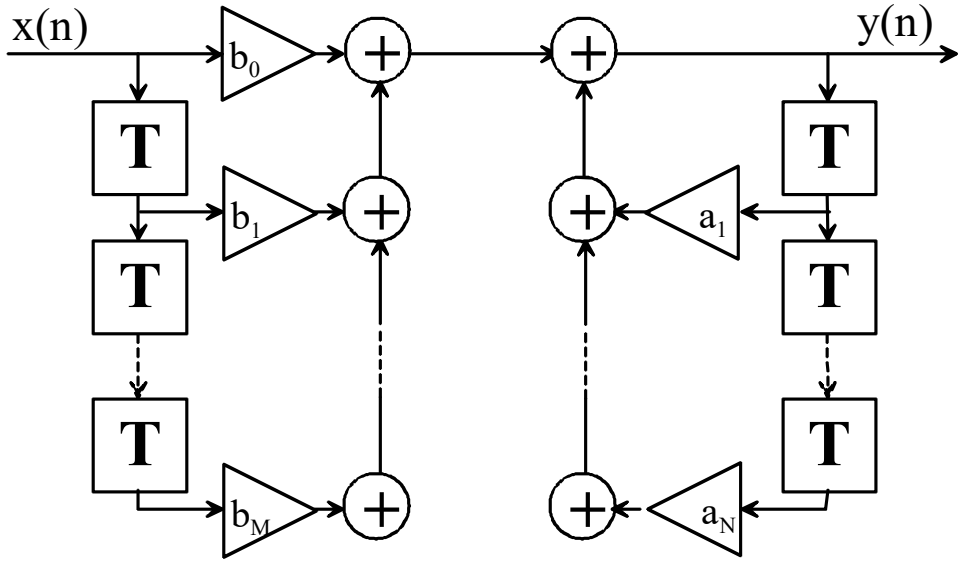
Drawbacks

- Only band-limited signals
- Rounding and quantization can cause errors
- Not too high frequencies

General Discrete-Time Filter

- Difference equation

$$y(n) = \sum_{k=1}^{N} a_k \cdot y(n-k) + \sum_{k=0}^{M} b_k \cdot x(n-k)$$



- FIR
  - $a_k = 0$, for all k
  - $b_k \neq 0$, for some k
  - Impulse response finite

$$h(n) = 0, \qquad n > M$$

- IIR
  - $a_k \neq 0$, for some k
  - Impulse response infinite

$$h(n) \neq 0, \qquad C < n < \infty$$

- By deciding $\{a_k\}$ and $\{b_k\}$ in an appropriate manner, a discrete-time filter is obtained with desired properties

Discrete Time Filters – FIR vs IIR

Finite Impulse Response (FIR)

- Always stable
- Enables linear phase response
- (Easy to Design)
- Require higher orders
  - Long delays
  - More computations

Infinite Impulse Response (IIR)

- Possible with sharper transitions
- Efficient implementation
  - Less delays
- Potential instability
- Phase distortion

Short-Time Fourier Transform Spectrogram

Frequency change over time

Short-Time Fourer Transform: Representing Time-Frequency Data

- **Fourier Transform**

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

  - Describes frequency contents of a signal
  - Describes the full time interval – no information on *when* certain frequencies occur

- **Short-Time Fourier Transform**
  - Time-frequency representation of a signal's frequency contents
  - Divide the signal into shorter signal by using a window function $w(t)$
    - e.g. rectangular, Hamming, Hann

$$X(t, \omega) = \int_{-\infty}^{\infty} x(\tau) w(t - \tau) e^{-j\omega\tau} d\tau$$

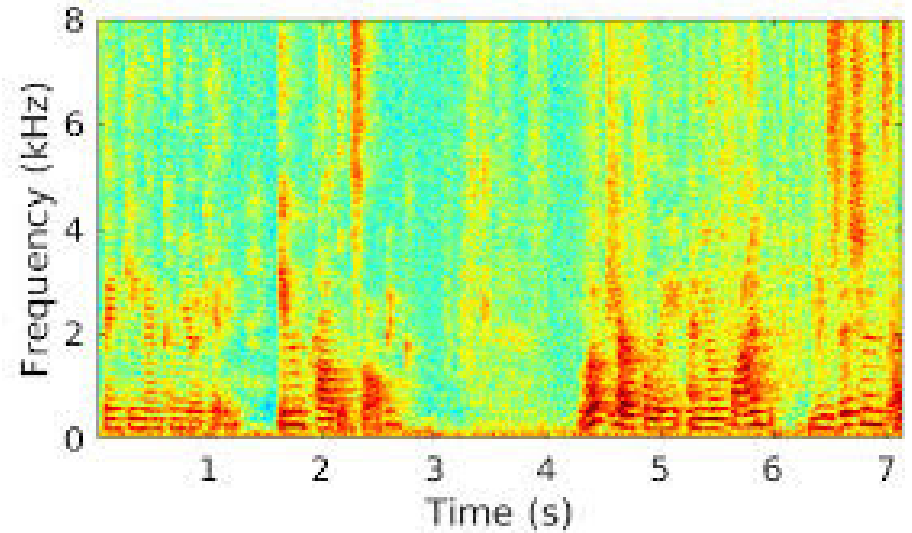
  - Large window: better frequency resolution, worse time resolution
  - Small window: better time resolution, worse frequency resolution

Spectrogram – The Frequency Contents over Time

- Spectrogram
  - Visual representation the signal's Power distribution over time

$$S(t, \omega) = |X(t, \omega)|^2$$

where $X(t, \omega)$ is the Short-Time Fourier Transform



Spectrogram – Applications

- **Speech Processing**
  - Spectrograms are used to visualize formants and understand how speech changes over time.
  - Example: Detecting vowels and consonants in speech signals.
- **Music Analysis**
  - Visualize harmonic structures and identify different instruments in a music piece.
- **Biomedical Signal Processing**
  - Used in EEG and ECG analysis to detect abnormal patterns in brain and heart activities.
- **Mechanical Vibration Analysis**
  - Detecting machinery faults based on frequency signatures over time.

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